EXCELLENCE

CLASS 11 - 2026-27

Physics | Vector 05

Vectors

Cross Product for Physics

Dot ⇒ $\vec{A} \cdot \vec{B} = |A||B|\cos\theta$

$= AB\cos\theta$

Cross ⇒ $\vec{A} \times \vec{B} = AB\sin\theta$

By - Excellence Programme

5 October 2026

Topics to be covered

01 Why physics needs A × B

02 Magnitude A B sin θ

03 Right-hand rule (direction)

04 Dot vs cross — memory

05 $\hat{i}\,\hat{j}\,\hat{k}$ cycle

06 xy-plane shortcut

07 Torque $\vec{\tau} = \vec{r} \times \vec{F}$ → Rotational motion

vector

08 Board practice

Physics, not maths chapter

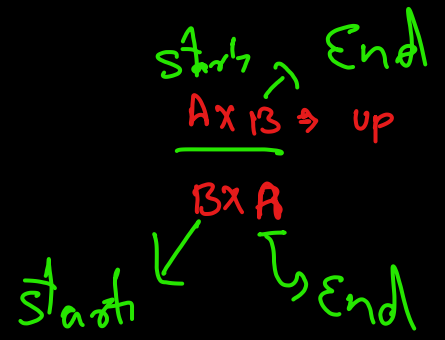
We only keep what Class 11 Physics actually uses.

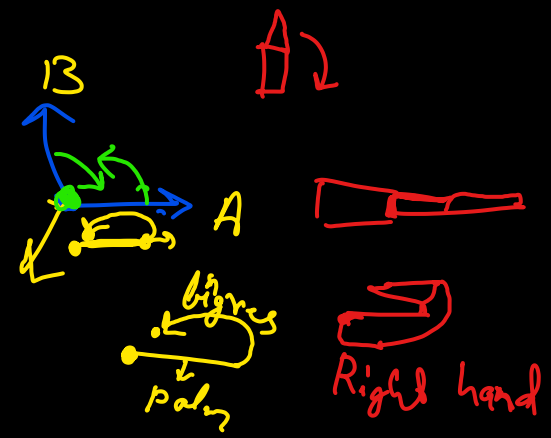
Keep: size A B sin θ, right-hand rule, $\hat{i}\hat{j}\hat{k}$, torque language.

Skip today: triangle/parallelogram area drills, long 3×3 algebra.

Cross product gives a vector perpendicular to both factors.

Later chapters use the same idea for torque and magnetic force.





On board

If clockwise ⇒ outward

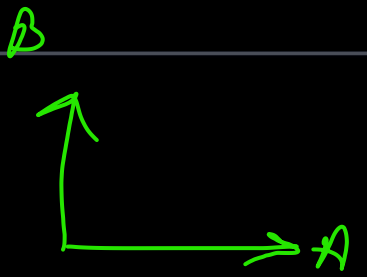
with ⇒ Inward

If Anticlockwise ⇒ outward

Why physics needs A × B

Dot product answered: how much along the same line?

Cross product answers: how much perpendicular turning effect?



A × B is perpendicular to both A and B

A × B = outward
B × A = inward

Think: A and B on a table. A × B stands straight up from the table.

If A and B are parallel, A × B = 0. No unique upright direction.

A · B = B · A ⇒ True

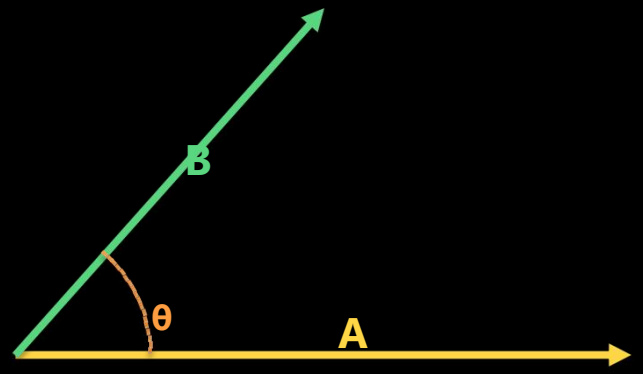
A × B = B × A ⇒ False

A × B = − B × A (dir^{n} opposite & magnitude same)

Magnitude first: A B sin θ

Only the perpendicular part of B counts. That part is B sin θ.

$$|A \times B| = A\,B \sin\theta$$



Memory

sin for cross, cos for dot.

Physics habit

Max at 90°. Zero at 0° or 180°.

$\theta = 0°$

$\sin\theta = \sin 0°$

$= 0$

Rough work / extra work

$a = 2\hat{i} + 4\hat{j}$

$b = 3\hat{i} + 3\hat{j}$

$a \cdot b \Rightarrow 6 + 12 = 18$

$b \cdot a \Rightarrow 18$

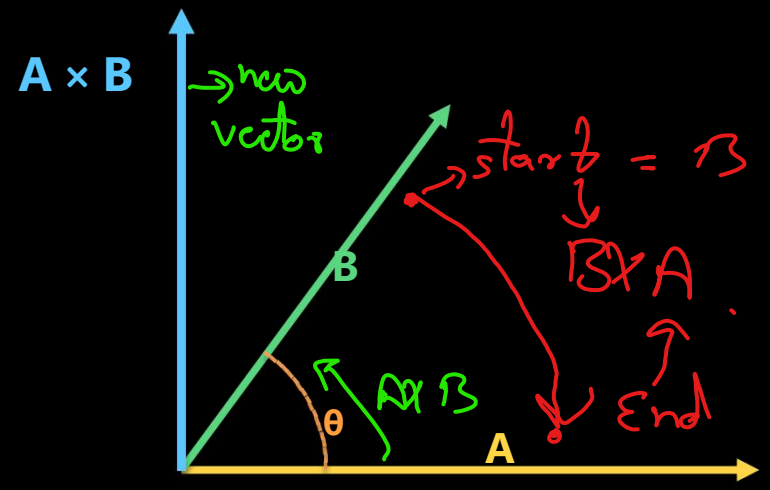
$a \cdot b = b \cdot a \Rightarrow$ True

$a \times b = b \times a \Rightarrow$ False $\Rightarrow$ $\boxed{a \times b = -b \times a}$

We add " − " minus here because dir. is opposition

Direction: right-hand rule

Fingers along A. Curl toward B through the smaller angle. Thumb = A × B.



$\vec{A} \cdot \vec{B}$ = value

$\vec{A} \times \vec{B}$ = (new vector)

$$A \times B = -(B \times A)$$

Order matters

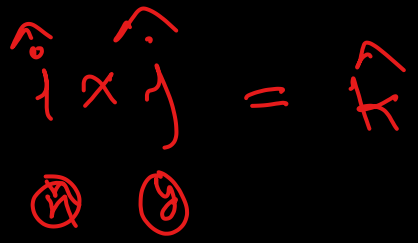
Swap the arrows → result flips.

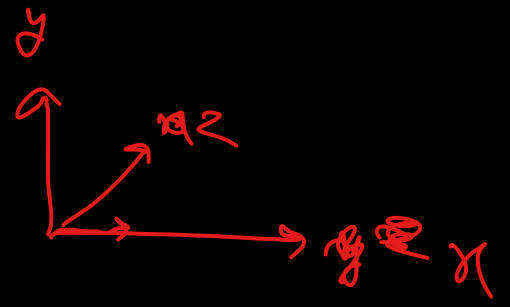
Right-hand rule is how physics fixes direction of torque.

Dot vs cross — physics memory

	Dot product	Cross product
Result	a number (scalar)	a vector
Size	A B cos θ	A B sin θ
Zero when	perpendicular	parallel
Order	A · B = B · A	A × B = − B × A
Used for	work, projection	torque, magnetic force

Cos asks how parallel. Sin asks how perpendicular.





$\hat{i}\,\hat{j}\,\hat{k}$ cycle (must know)

$$\hat{i} \times \hat{j} = \hat{k}\,,\ \hat{j} \times \hat{k} = \hat{i}\,,\ \hat{k} \times \hat{i} = \hat{j}$$

$\hat{i} \times \hat{k} = -\hat{j}$

$\hat{a} \times \hat{b} = \hat{j}$

Forward in the cycle i → j → k → i: positive.

$\hat{b} \times \hat{a} = -\hat{j}$

Backward: minus sign. Example: $\hat{j} \times \hat{i} = -\hat{k}$.

$$\hat{i} \times \hat{i} = 0\,,\ \hat{j} \times \hat{j} = 0\,,\ \hat{k} \times \hat{k} = 0$$

Same unit vector crossed with itself is always zero.

$\hat{i} \times \hat{i} = 0$ (why?)

$\theta = 0$

So $\sin\theta = 0$

xy-plane shortcut (enough for now)

Most Class 11 board problems stay in the plane.

If $A = A_x \hat{i} + A_y \hat{j}$ and $B = B_x \hat{i} + B_y \hat{j}$:

$$\mathbf{A \times B = (A_x B_y - A_y B_x)\,\hat{k}}$$

Result points along $\pm \hat{k}$ only — out of the plane.

Do not start heavy 3×3 algebra unless a problem forces 3D.

Torque is a cross product

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$$

So $|\tau| = r\,F \sin\theta$. Same sin pattern as $|A \times B|$.

Direction of τ: right-hand rule with fingers $r \rightarrow F$.

Max torque

Force perpendicular to r.

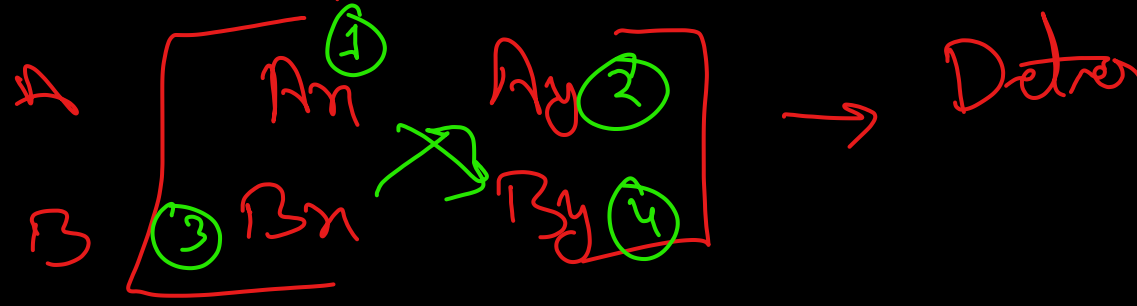
Zero torque

Force parallel to r ($\theta = 0°$ or $180°$).

Rough work / extra work

$\vec{A} = A_x \hat{i} + A_y \hat{j}$

$\vec{B} = B_x \hat{i} + B_y \hat{j}$



$\rightarrow$ Determinant $\Rightarrow$ $A_x B_y - A_y B_x$

$1 \times 4 - 2 \times 3$

Where else this idea appears

1. Torque: turning effect of a force about a point.

2. Later magnetism: $F = q\,(v \times B)$ uses the same rule.

3. Angular momentum language: $L = r \times p$ (same shape).

Learn the tool once. Physics reuses it many times.

Today: master size + direction. Applications grow later.

Find |A × B|

Solve It

A = 3 and B = 4. Angle between them is 30°. Find |A × B|.

$4 \times 3 \times \sin 30$

$4 \times 3 \times \frac{1}{2} = 2 \times 3 \Rightarrow 6$

Solution: Find |A × B|

Step 1: $|A \times B| = A\,B \sin\theta = 3 \times 4 \times \sin 30°$

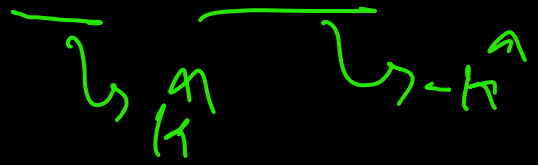
Step 2: $\sin 30° = 1/2$, so $|A \times B| = 6$

$|A \times B| = 6$.

$\hat{i} \times \hat{j}$ and $\hat{j} \times \hat{i}$

Solve

Find $\hat{i} \times \hat{j}$ and $\hat{j} \times \hat{i}$.



Solution: $\hat{i} \times \hat{j}$ and $\hat{j} \times \hat{i}$

Step 1: $\hat{i} \times \hat{j} = \hat{k}$

Step 2: $\hat{j} \times \hat{i} = -(\hat{i} \times \hat{j}) = -\hat{k}$

$\hat{k}$ and $-\hat{k}$. Order flips the sign.

Rough work / extra work

Cross product in the xy-plane

Determinant ⇒

A = 2 î + 3 ĵ and B = 4 î − ĵ. Find A × B.

$$\begin{vmatrix} 2 & 3 \\ 4 & -1 \end{vmatrix} \hat{k}$$

$= -2 \times 1 - 3 \times 4$

$= -2 - 12$

$= -14\hat{k}$

Solution: Cross product in the xy-plane

Step 1: $\vec{A} \times \vec{B} = (A_x B_y - A_y B_x)\,\hat{k}$

Step 2: $= (2)(-1) - (3)(4) = -2 - 12 = -14$

$\vec{A} \times \vec{B} = -14\,\hat{k}$.

When is A × B zero?

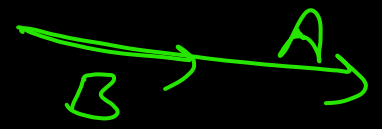
Solve ff

A = 2 î + 4 ĵ and B = î + 2 ĵ. Find A × B. What does that mean physically?

$\vec{A} \times \vec{B} = 0$

↳ mean $\theta = 0$

↳ Then A & B // to each other



Solution: When is A × B zero?

Step 1: $A \times B = (2\times2 - 4\times1)\,\hat{k} = 0$

Step 2: B is parallel to A, so there is no perpendicular turning part

$A \times B = 0$. Parallel vectors → zero cross product.

Torque magnitude

$\vec{\tau} = \vec{r} \times \vec{F} \Rightarrow |\tau|$

A force F = 10 N acts at r = 0.4 m from the pivot. Angle between r and F is 90°. Find $|\tau|$.

$|\tau| = |r||F| \sin\theta$

$= 0.4 \times 10 \times \sin 90^\circ$

$= 4$

sign → Torque

Solution: Torque magnitude

Step 1: $|\tau| = r F \sin\theta = 0.4 \times 10 \times \sin 90^\circ$

Step 2: $\sin 90^\circ = 1$, so $|\tau| = 4$ N m

$|\tau| = 4$ N m.

Rough work / extra work

When is torque zero?

Solve It

Same r = 0.4 m and F = 10 N, but force is along the line of r (θ = 0°). Find |τ|.

$\tau = rF \sin 0$

$\tau = rF \times 0$

$|\vec{\tau}| = 0$

Solution: When is torque zero?

Step 1: $|\tau| = r F \sin 0° = 0$

Step 2: Force through the pivot line gives no turning effect

$|\tau| = 0$. Parallel → no torque.

Maximum |A × B|

max (sin θ)

|A| = 6 and |B| = 5. What is the maximum possible |A × B|?

30

6 x 5 x 1

max (AB sin θ)

AB max (sin θ) = 1

6 x 5 x 1

30

Solution: Maximum |A × B|

Step 1: $|A \times B| = A\,B \sin\theta \le A\,B$

Step 2: Maximum when $\theta = 90°$ (fully perpendicular)

Maximum $|A \times B| = 30$.

Rough work / extra work

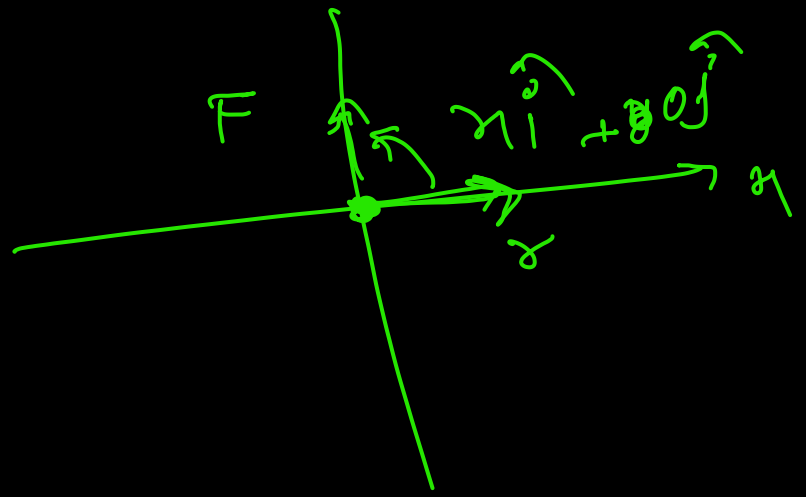
Direction of torque

$\tau = ?$

r is along +x (î) and F is along +y (ĵ). Direction of τ = r × F is

1. +x
2. +y
3. +z (k̂)
4. −z (−k̂)

$\vec{r} \times \vec{F} =$



Solution: Direction of torque

Step 1: $\hat{i} \times \hat{j} = \hat{k}$

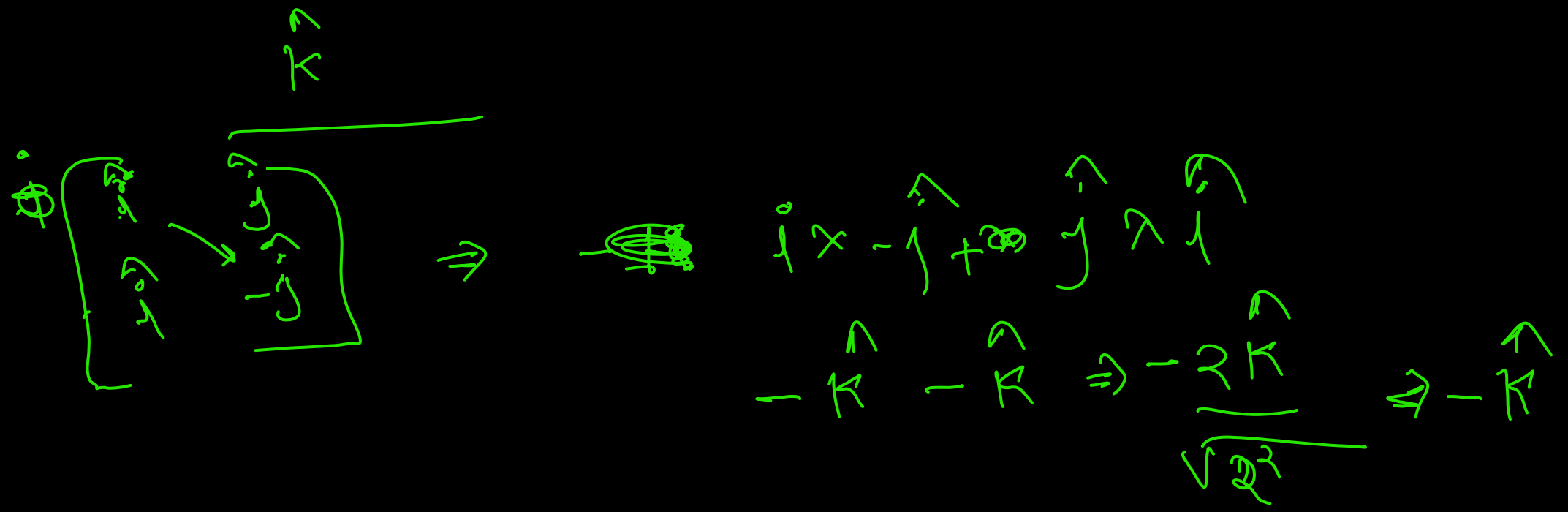
Step 2: Right-hand rule: fingers x → y, thumb out of page (+z)

Answer: +z ($\hat{k}$).

Unit vector along A × B

Solve It

A = î + ĵ and B = î − ĵ. Find a unit vector along A × B.



Solution: Unit vector along A × B

Step 1: $\vec{A} \times \vec{B} = (1\times(-1) - 1\times1)\,\hat{k} = -2\,\hat{k}$

Step 2: Unit vector = $(-2\,\hat{k})/2 = -\,\hat{k}$

Unit vector = $-\,\hat{k}$.

If A · B = 0, what about |A × B|?

Solve If

If A and B are perpendicular and non-zero, |A × B| equals

1. 0
2. A B
3. A + B
4. A / B

$A \cdot B = 0$

$AB\cos\theta = 0$

$\cos\theta = 0$

$\theta = 90°$

$|A \times B| = AB\sin\theta$

$= AB \times 1$

$= AB$

Solution: If A · B = 0, what about |A × B|?

Step 1: $\theta = 90° \Rightarrow \sin\theta = 1 \Rightarrow |A \times B| = A\,B$

Answer: A B. (Same idea as max torque.)

Rough work / extra work

Vector 05 lock lines

$|A \times B| = A\,B \sin\theta$. Max at 90°. Zero if parallel.

Direction by right-hand rule. $A \times B = -B \times A$.

$\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$

In the plane: $A \times B = (A_x B_y - A_y B_x)\,\hat{k}$

Physics link: $\tau = r \times F$, so $|\tau| = r\,F \sin\theta$.

Practice

1. Find $|A \times B|$ if $A = 5$, $B = 8$, $\theta = 90°$.

2. Write $\hat{i} \times \hat{j}$, $\hat{j} \times \hat{k}$, $\hat{k} \times \hat{i}$, and $\hat{j} \times \hat{i}$.

3. $A = 3\,\hat{i} + \hat{j}$, $B = 2\,\hat{i} + 4\,\hat{j}$. Find $A \times B$.

4. $r = 0.5$ m, $F = 20$ N, $\theta = 30°$. Find $|\tau|$.

5. Force along the rod through the pivot: what is torque? Why?

6. r along $\hat{j}$, F along $-\hat{i}$. Direction of τ?

Dot uses cos. Cross uses sin. Torque uses the same sin.

Vectors Ends.

Tommorow unit 4 dim